

Data Analytics

DECISION TREE

FINAL PROJECT

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PLAN AND PREDICT

Target (Y):

Predicting diabetes risk is valuable because early detection allows for lifestyle interventions before the disease fully develops. This model could help find high-risk patients using routine measurements, no lab-intensive testing required.

Predictors (X):

Glucose - Blood Sugar level is the most direct biological marker of diabetes

BMI - Obesity is a leading risk factor for Type 2 diabetes

Age - Diabetes risk increases with age

Pregnancies - Gestational diabetes history raises long-term risk

Blood Pressure - Hypertension frequently co-occurs with diabetes

Hypothesis:

Most Important Factor: Glucose: High blood glucose is how diabetes is diagnosed directly; the model should mirror this

Second Most Important Factors: BMI: After glucose, weight status is the strongest modifiable risk factor and should drive the second split.

BASELINE TREE

Decision Tree (max_depth=2) — Test Accuracy: 77.27%

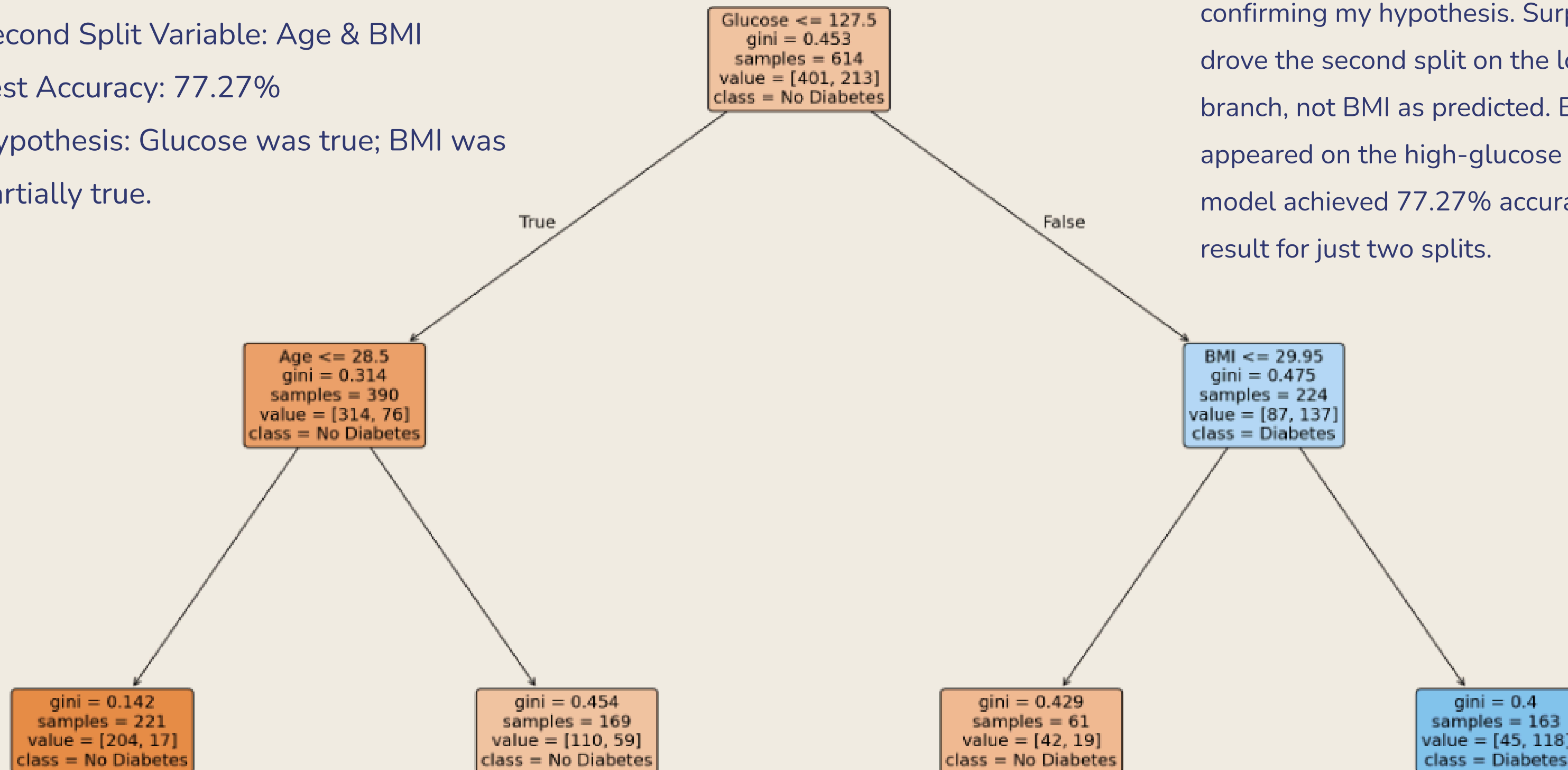
First Split Variable: Glucose ≤ 127.5

Second Split Variable: Age & BMI

Test Accuracy: 77.27%

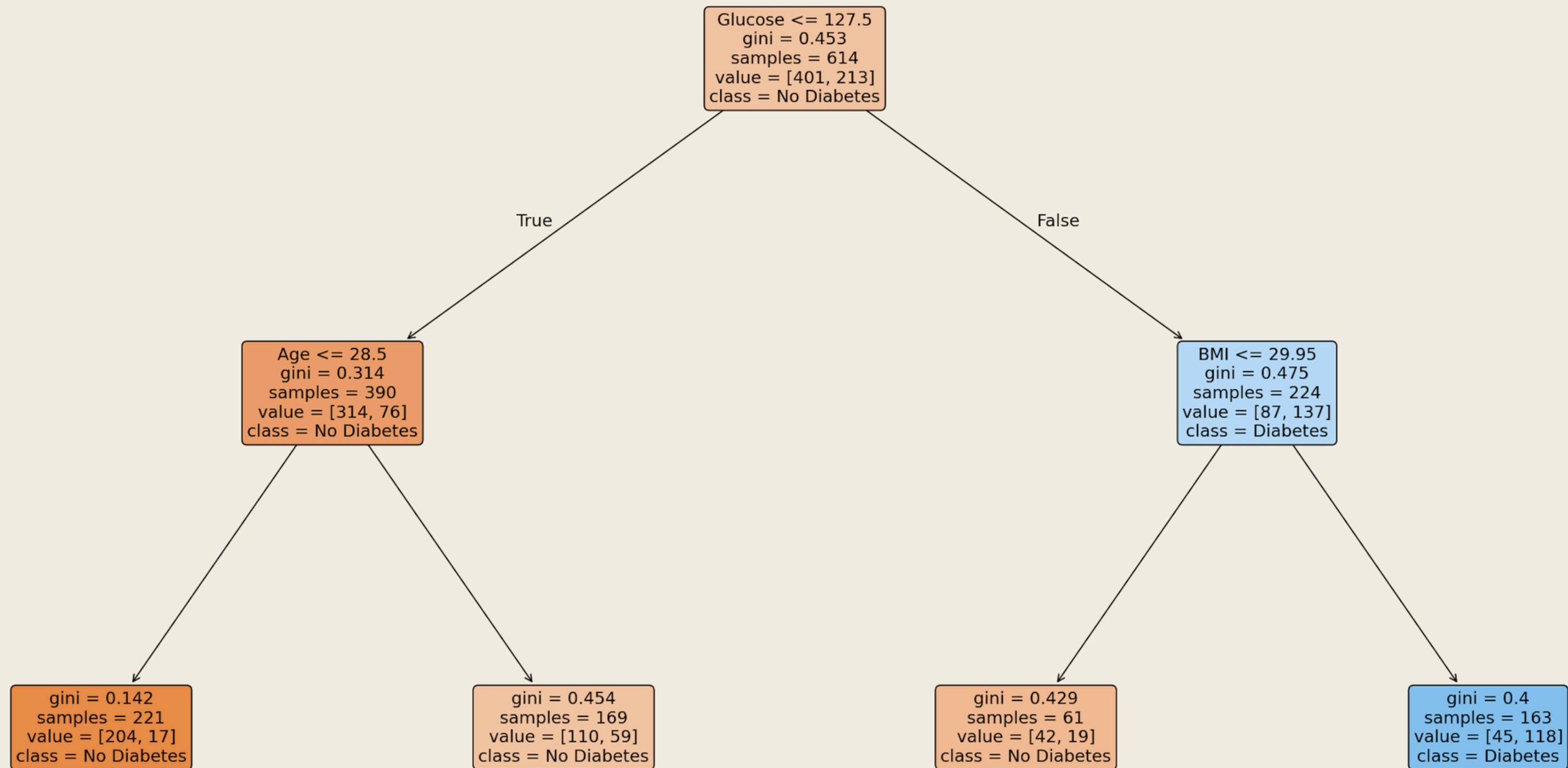
Hypothesis: Glucose was true; BMI was partially true.

The model first splits on Glucose (≤ 127.5), confirming my hypothesis. Surprisingly, Age drove the second split on the low-glucose branch, not BMI as predicted. BMI still appeared on the high-glucose branch. The model achieved 77.27% accuracy, a solid result for just two splits.

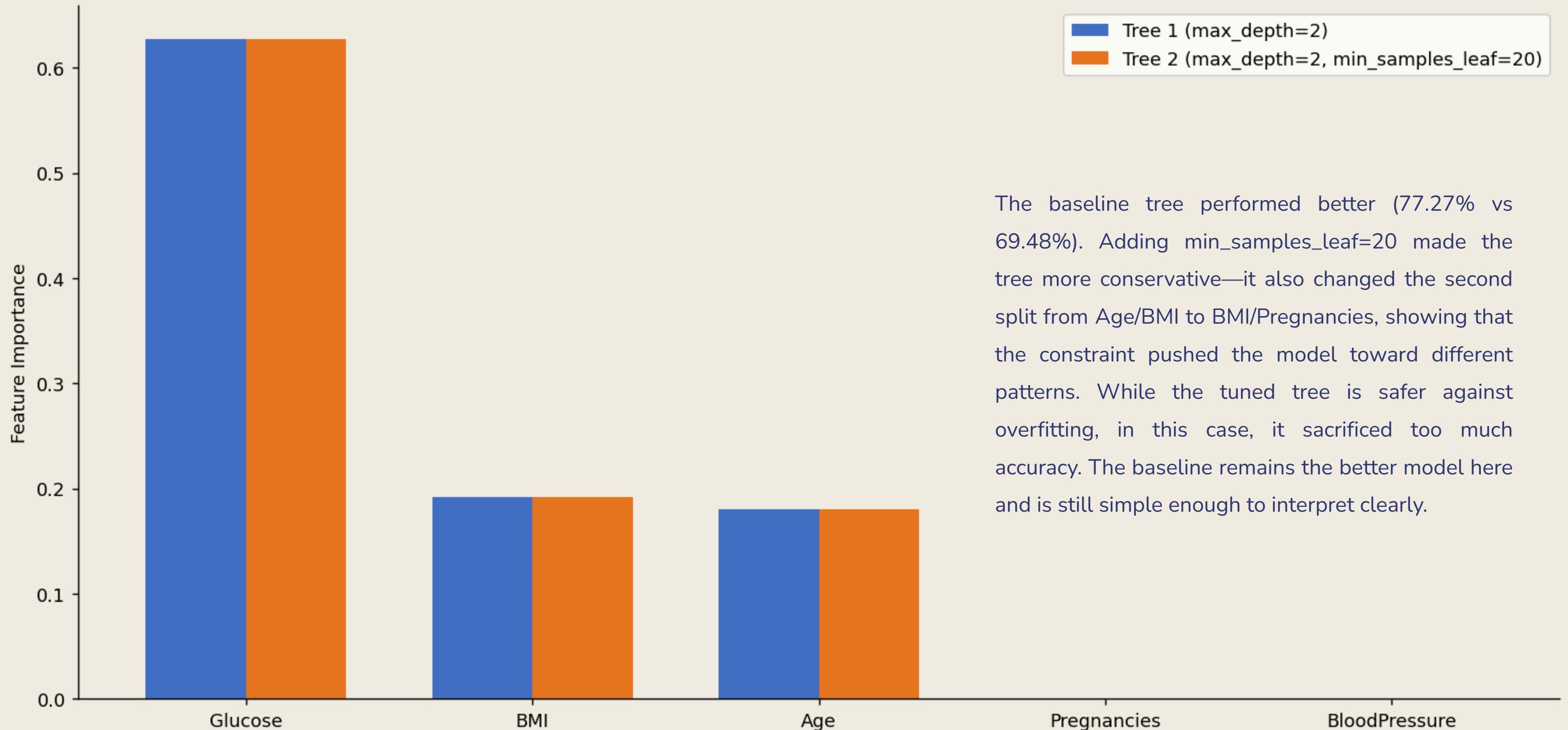


BASELINE TREE

Decision Tree (max_depth=2, min_samples_leaf=20) — Test Accuracy: 77.27%



Feature Importances: Tree 1 vs. Tree 2



The baseline tree performed better (77.27% vs 69.48%). Adding `min_samples_leaf=20` made the tree more conservative—it also changed the second split from Age/BMI to BMI/Pregnancies, showing that the constraint pushed the model toward different patterns. While the tuned tree is safer against overfitting, in this case, it sacrificed too much accuracy. The baseline remains the better model here and is still simple enough to interpret clearly.